



VP AIR TEST NEET (2025-26)

PART TEST - 06

DROPPER

DURATION: 180 Minutes

Date: 30-11-2025

M. MARKS: 720

ANSWER KEY

1. (1)	46. (3)	91. (3)	136. (2)
2. (1)	47. (3)	92. (2)	137. (2)
3. (2)	48. (2)	93. (1)	138. (3)
4. (1)	49. (1)	94. (4)	139. (1)
5. (2)	50. (4)	95. (3)	140. (1)
6. (2)	51. (3)	96. (4)	141. (4)
7. (3)	52. (2)	97. (1)	142. (2)
8. (4)	53. (3)	98. (2)	143. (1)
9. (4)	54. (1)	99. (3)	144. (3)
10. (4)	55. (4)	100. (2)	145. (2)
11. (1)	56. (4)	101. (1)	146. (4)
12. (2)	57. (3)	102. (2)	147. (3)
13. (3)	58. (1)	103. (3)	148. (4)
14. (1)	59. (2)	104. (4)	149. (3)
15. (1)	60. (1)	105. (2)	150. (4)
16. (2)	61. (3)	106. (1)	151. (4)
17. (3)	62. (3)	107. (1)	152. (3)
18. (3)	63. (1)	108. (4)	153. (2)
19. (1)	64. (2)	109. (3)	154. (3)
20. (2)	65. (2)	110. (2)	155. (2)
21. (3)	66. (1)	111. (2)	156. (3)
22. (3)	67. (3)	112. (2)	157. (1)
23. (3)	68. (3)	113. (4)	158. (3)
24. (1)	69. (1)	114. (1)	159. (2)
25. (2)	70. (3)	115. (3)	160. (1)
26. (2)	71. (1)	116. (2)	161. (4)
27. (1)	72. (4)	117. (3)	162. (4)
28. (3)	73. (1)	118. (3)	163. (2)
29. (4)	74. (2)	119. (2)	164. (2)
30. (3)	75. (4)	120. (2)	165. (1)
31. (4)	76. (1)	121. (1)	166. (1)
32. (2)	77. (3)	122. (3)	167. (3)
33. (1)	78. (1)	123. (1)	168. (2)
34. (2)	79. (3)	124. (3)	169. (3)
35. (1)	80. (4)	125. (2)	170. (3)
36. (1)	81. (1)	126. (3)	171. (3)
37. (3)	82. (3)	127. (1)	172. (2)
38. (2)	83. (3)	128. (4)	173. (2)
39. (1)	84. (2)	129. (3)	174. (2)
40. (1)	85. (1)	130. (2)	175. (1)
41. (4)	86. (2)	131. (3)	176. (3)
42. (2)	87. (3)	132. (1)	177. (3)
43. (2)	88. (4)	133. (1)	178. (1)
44. (2)	89. (1)	134. (2)	179. (2)
45. (3)	90. (4)	135. (4)	180. (2)

HINTS & SOLUTIONS**1. (1)**

$$\text{Let total energy} = E = \frac{1}{2}KA^2$$

$$U = \frac{1}{2}K\left(\frac{A}{3}\right)^2 = \frac{KA^2}{2 \times 9} = \frac{E}{9}$$

$$KE = E - \frac{E}{9} = \frac{8E}{9}$$

$$\text{Ratio} = \frac{\text{Total Energy}}{KE} = \frac{E}{\frac{8E}{9}} = \frac{9}{8}$$

$$x = 9$$

(NCERT Ed. 2025-26, 11th, Page No. 268)**2. (1)**

$$y = 5 \sin \frac{2\pi x}{3} \cos 40\pi t \dots (i)$$

The standard equation of a stationary wave is

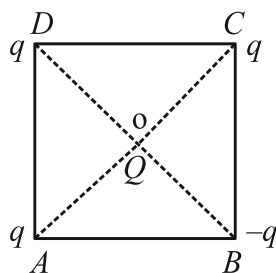
$$y = 2a \sin kx \cos \omega t \dots (ii)$$

$$\text{Comparing (i) and (ii), we get } k = \frac{2\pi}{3}$$

$$\text{or } \frac{2\pi}{\lambda} = \frac{2\pi}{3} \text{ or } \lambda = 3 \text{ cm}$$

Separation between two adjacent nodes is

$$\frac{\lambda}{2} = \frac{3}{2} = 1.5 \text{ cm}$$

(NCERT Ed. 2025-26, 11th, Page No. 290)**3. (2)**

$$\vec{F}_{net} = \vec{F}_A + \vec{F}_B + \vec{F}_C + \vec{F}_D$$

$$= \frac{2kqQ}{r^2} \text{ towards } \overline{OB}$$

(NCERT Ed. 2025-26, 12th, Page No. 12)**4. (1)**

With humidity, the speed of sound increases.

(NCERT Ed. 2025-26, 11th, Page No. 287)**5. (2)**

$$\text{Total potential energy} = 0.04 \text{ J}$$

$$\text{Potential energy at mean position} = 0.01 \text{ J}$$

$$\text{Maximum kinetic energy} = (0.04 - 0.01) = 0.03 \text{ J}$$

$$0.03 = \frac{1}{2}ka^2$$

$$0.03 = \frac{1}{2} \times k \times \left(\frac{20}{1000}\right)^2$$

$$k = 0.06 \times 2500 \text{ N/m} = 150 \text{ N/m}$$

(NCERT Ed. 2025-26, 11th, Page No. 269)**6. (2)**

Charge is always quantized in nature.

So, $q = ne$ where n is integer.

Total net charge of an isolated system is always conserved.

(NCERT Ed. 2025-26, 12th, Page No. 5)**7. (3)**

$$E = \frac{KQ}{R^2} = 3 \times 10^6$$

$$\Rightarrow \frac{9 \times 10^9 \times Q \times 4}{25} = 3 \times 10^6$$

$$\Rightarrow Q = \frac{25}{12} \times 10^{-3} \approx 2 \times 10^{-3} \text{ C}$$

(NCERT Ed. 2025-26, 12th, Page No. 35)**8. (4)**

$$\text{speed: } v = \sqrt{\frac{\gamma RT}{M}}$$

When temperature is kept constant speed does not depend on pressure.

(NCERT Ed. 2025-26, 11th, Page No. 287)**9. (4)**

$$\tau = pE \sin \theta = 1 \times 10^5 p = 4$$

$$\Rightarrow p = 4 \times 10^{-5} = (q)(2 \times 10^{-2})$$

$$\Rightarrow q = 2 \times 10^{-3} = 2 \text{ mC.}$$

(NCERT Ed. 2025-26, 12th, Page No. 27)**10. (4)**

$$\frac{v}{L_O} = \frac{3v}{4L} \therefore L_O = \frac{4L}{3}$$

(NCERT Ed. 2025-26, 11th, Page No. 292)**11. (1)**The net force ($T - mg$) is toward the center of the circle, which provides the required centripetal acceleration.**(NCERT Ed. 2025-26, 11th, Page No. 270)**

12. (2)

$$F = K(1) \left(\frac{1}{1} + \frac{1}{4} + \frac{1}{16} + \dots \right) \times 10^{-6}$$

$$= K \left[\frac{4}{3} \right] \times 10^{-6} = \frac{9 \times 10^9 \times 4 \times 10^{-6}}{3}$$

$$= 12000 \text{ N.}$$

(NCERT Ed. 2025-26, 12th, Page No. 12)

13. (3)

Total electric flux through closed surface is $\frac{q}{\epsilon_0}$

So electric flux through each face = $\frac{q/\epsilon_0}{6}$

$$= \frac{q}{6\epsilon_0}$$

(NCERT Ed. 2025-26, 12th, Page No. 31)

14. (1)

$$T = 2\pi \sqrt{\frac{m}{k}}$$

The time period does not change only the mean position changes as, buoyant force will shift the block a little upwards from its initial mean position.

(NCERT Ed. 2025-26, 11th, Page No. 268)

15. (1)

$$100 = \omega A$$

$$100 = 10\pi A \Rightarrow A = \frac{10}{\pi} \text{ m}$$

(NCERT Ed. 2025-26, 11th, Page No. 266)

16. (2)

$$F = \frac{kq_1q_2}{r^2} \text{ coulomb's law}$$

$$\phi = \frac{q}{\epsilon_0} \text{ Gauss's law}$$

Principle of superposition involves vector sum of forces.

Quantisation of charges $\Rightarrow$ discrete nature of charges.

(NCERT Ed. 2025-26, 12th, Page No. 37)

17. (3)

$$v = \frac{1}{2} \sqrt{(25 - x^2)}$$

On comparing

$$\omega = \frac{1}{2} = \frac{2\pi}{T}$$

$$\therefore T = 4\pi$$

(NCERT Ed. 2025-26, 11th, Page No. 268)

18. (3)

Electric field $\vec{E}$ inside the uniformly charged non conducting solid sphere at distance x from its centre is.

$$|\vec{E}| = \frac{kQx}{r^3}$$

(NCERT Ed. 2025-26, 12th, Page No. 37)

19. (1)

$$2 = \frac{v}{2L} - \frac{v}{4L} = \frac{v}{4L}$$

$$f' = \frac{v}{L} - \frac{v}{8L} = \frac{7v}{8L} = \frac{7 \times 8}{8} = 7$$

(NCERT Ed. 2025-26, 11th, Page No. 293)

20. (2)

$$\phi_{net} = \frac{Q}{\epsilon_0} = \phi_{cs} + 2\phi_{cone}$$

$$\Rightarrow \phi_{cone} = \frac{1}{2} \left(\frac{Q}{\epsilon_0} - \phi \right)$$

(NCERT Ed. 2025-26, 12th, Page No. 30)

21. (3)

$$\frac{F_1}{F_2} = \frac{25}{5} = \frac{r_2^2}{r_1^2}$$

$$r_2^2 = 5r_1^2$$

$$r_2 = \sqrt{5}r_1$$

(NCERT Ed. 2025-26, 12th, Page No. 7)

22. (3)

$$X = A \sin(\omega t + \phi)$$

$$0 = A \sin(\omega t + \phi)$$

$$\sin(\omega t + \phi) = 0$$

$$\omega t + \phi = 0^\circ \text{ or } 180^\circ$$

Since the two particles are going in opposite direction, the phase for one is 0° and for another is 180° .

So the phase difference is $180^\circ - 0^\circ = 180^\circ$.

(NCERT Ed. 2025-26, 11th, Page No. 263)

23. (3)

$$T \propto \sqrt{L}, \sqrt{\frac{L_1}{L_2}} = \frac{T_1}{T_2} = \frac{7}{9}$$

$$9 \times \sqrt{L_1} = 7 \sqrt{L_2} \Rightarrow \frac{L_1}{L_2} = \frac{49}{81}$$

(NCERT Ed. 2025-26, 11th, Page No. 271)

24. (1)

Since the sphere is conducting charge resides on the surface only.

(NCERT Ed. 2025-26, 12th, Page No. 3)

25. (2)

$$m_{\text{reduced}} = \frac{m \cdot m}{m + m} = \frac{m}{2}$$

$$T = 2\pi\sqrt{\frac{m_{\text{reduced}}}{k}} = 2\pi\sqrt{\frac{m}{2k}}$$

(NCERT Ed. 2025-26, 11th, Page No. 276)

26. (2)

Torque on electric dipole is,

$$\vec{\tau} = \vec{p} \times \vec{E}$$

(NCERT Ed. 2025-26, 12th, Page No. 27)

27. (1)

In transverse mechanical waves, the constituents of the medium oscillates perpendicular to the direction of wave propagation.

In longitudinal waves, the constituents of the medium oscillates along the direction of wave propagation.

Both the statements are correct.

(NCERT Ed. 2025-26, 11th, Page No. 280)

28. (3)

$$f \propto \frac{1}{\sqrt{\mu}} \propto \frac{1}{\sqrt{r^2}}, [\mu = \rho\pi r^2]$$

$$f \propto \frac{1}{r}$$

(NCERT Ed. 2025-26, 11th, Page No. 285)

29. (4)

$$\frac{(k)(100q)}{x^2} = \frac{k(q)}{(44-x)^2}$$

$$\Rightarrow 10 = \frac{x}{44-x}$$

$$440 - 10x = x$$

$$440 = 11x$$

$$x = 40 \text{ cm}$$

(NCERT Ed. 2025-26, 12th, Page No. 12)

30. (3)

$$x_1 = A \sin(\omega t) = A \cos\left(\omega t - \frac{\pi}{2}\right)$$

$$x_2 = A \cos\left(\omega t + \frac{\pi}{4}\right)$$

$$\text{Phase difference } \Delta\phi = \omega t + \frac{\pi}{4} - \omega t + \frac{\pi}{2} = \frac{3\pi}{4}$$

(NCERT Ed. 2025-26, 11th, Page No. 263)

31. (4)

Inside the shell, electric field is zero. Both net force and net torque on the dipole is zero.

(NCERT Ed. 2025-26, 12th, Page No. 36)

32. (2)

$$|f_A - f_B| = 6 \text{ Hz}$$

When tension in B increases f_B increases.

As the beats frequency is increasing

$$\text{Therefore } f_B - f_A = 6$$

$$f_B = 6 + 530 = 536 \text{ Hz}$$

(NCERT Ed. 2025-26, 11th, Page No. 294)

33. (1)

$$q_f = \frac{40 - 20}{2} = 10 \mu\text{C}$$

$$\frac{F_1}{F_2} = \frac{(40)(20)}{(10)(10)} = \frac{8}{1}$$

(NCERT Ed. 2025-26, 12th, Page No. 7)

34. (2)

$$y = y_0 \sin\left(\frac{2\pi vt}{\lambda} - \frac{2\pi x}{\lambda}\right)$$

Wave velocity = v

Particle velocity

$$= \left(\frac{\partial y}{\partial t}\right)_x = y_0 \cos\left(\frac{2\pi vt}{\lambda} - \frac{2\pi x}{\lambda}\right) \left(\frac{2\pi v}{\lambda}\right)$$

$$\left[\left(\frac{\partial y}{\partial t}\right)_x\right]_{\text{max}} = \frac{2\pi v}{\lambda} y_0$$

$$\text{If } v_P = 3v_W$$

$$\frac{2\pi v y_0}{\lambda} = 3v$$

$$\lambda = \frac{2\pi y_0}{3}$$

(NCERT Ed. 2025-26, 11th, Page No. 281)

35. (1)

The given kinetic energy is $K = K_0 \cos^2 \omega t$.

The maximum value of $\cos^2 \omega t$ is 1.

Therefore, $K_{\text{max}} = K_0$.

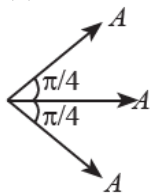
Energy Conservation in SHM:

In simple harmonic motion:

$$E = K_{\text{max}} = U_{\text{max}} = K_0$$

(NCERT Ed. 2025-26, 11th, Page No. 268)

36. (1)



$$A_{\text{res}} = (\sqrt{2} + 1)A$$

$$I_{\text{res}} = (\sqrt{2} + 1)^2 I_0 = (3 + 2\sqrt{2}) I_0 = 5.8 I_0$$

(NCERT Ed. 2025-26, 11th, Page No. 287)

37. (3)

At $t = \frac{3T}{4}$, particle is at mean position

$\Rightarrow$ Force = 0

At $t = T$, particle is at extreme position

$\Rightarrow$ Acceleration is maximum

At $t = \frac{T}{4}$, particle is at mean position

$\Rightarrow$ Speed is maximum

At $t = \frac{T}{2}$, particle is at extreme position

$\Rightarrow$ Speed = 0 $\Rightarrow$ K.E. = 0

$\Rightarrow$ P.E. is maximum

(NCERT Ed. 2025-26, 11th, Page No. 262)

38. (2)

On cutting each part has spring constant $4K$

$$T \propto \frac{1}{\sqrt{K}} \Rightarrow T' = \frac{T}{2}$$

(NCERT Ed. 2025-26, 11th, Page No. 268)

39. (1)

Direction of Wave:

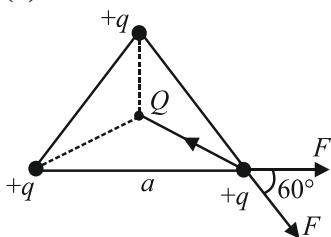
The argument of sine is $(10\pi x + 15\pi t)$. Since both x and t terms have the same sign (positive), the wave travels in the negative x -direction.

Wave Velocity (v):

$$v = \frac{15\pi}{10\pi} = 1.5 \text{ m/s}$$

(NCERT Ed. 2025-26, 11th, Page No. 281)

40. (1)



Q should be opposite in nature to q

$$\sqrt{3}F + \frac{kQq}{a^2}(3) = 0$$

$$\frac{\sqrt{3}kq^2}{a^2} + \frac{3kQq}{a^2} = 0$$

$$\Rightarrow Q = -\frac{q}{\sqrt{3}}$$

(NCERT Ed. 2025-26, 12th, Page No. 13)

41. (4)

The restoring torque is provided by a component of weight of the bob.

(NCERT Ed. 2025-26, 11th, Page No. 271)

42. (2)

$$v_p = -(\text{slope}) v_{\text{wave}}$$

As v_{wave} is -ve

$\therefore$ Wherever slope is positive particles are going up and vice versa.

Particles at D, E, F have downward velocity.

(NCERT Ed. 2025-26, 11th, Page No. 281)

43. (2)

Fundamental Open (A): Basic formula

$$f = \frac{v}{2L} \rightarrow \text{(III)}$$

2nd Harmonic Open (B): $\rightarrow$ (I)

1st Overtone Closed (C): This is the 3rd Harmonic

$$= 3\left(\frac{v}{4L}\right) = \frac{3v}{4L} \rightarrow \text{(IV)}$$

Fundamental Closed (D): Basic formula

$$f = \frac{v}{4L} \rightarrow \text{(II)}$$

(NCERT Ed. 2025-26, 11th, Page No. 291)

44. (2)

Velocity of particle = $-v_{\text{wave}} \times \text{slope}$

At $x_1 = 1.5$ m, slope is positive therefore it is going down.

At $x = 2.5$ m, slope is negative therefore it is going up.

(NCERT Ed. 2025-26, 11th, Page No. 281)

45. (3)

Fundamental frequency = $490 - 420 = 70$ Hz

$$70 = \frac{1}{2\ell} \sqrt{\frac{T}{\mu}}$$

($\because \mu$ is mass per unit length and T is tension in the wire)

$$\Rightarrow 70 = \frac{1}{2\ell} \sqrt{\frac{540}{6 \times 10^{-3}}}$$

$$\Rightarrow \ell = \frac{1}{2 \times 70} \sqrt{90 \times 10^3} = \frac{300}{140} = 2.14 \text{ m}$$

(NCERT Ed. 2025-26, 11th, Page No. 291)

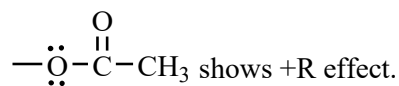
46. (3)

Metamers are the isomers having same molecular formula and same functional group but different alkyl or aryl groups on either side of functional group.

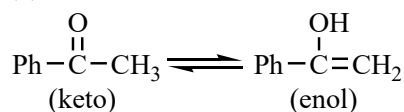
(NCERT Ed. 2025-26 11th Part-II Page No. 271)

47. (3)

Due to presence of lone pair on O-atom

(NCERT Ed. 2025-26 11th Part-II Page No. 276)

48. (2)

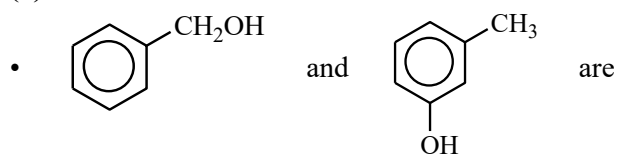
(NCERT Ed. 2025-26 11th Part-II Page No. 317)

49. (1)

Due to complete octet of N, NH_4^+ does not behave as an electrophile.

(NCERT Ed. 2025-26 11th Part-II Page No. 273)

50. (4)



functional isomers as aliphatic alcohols and phenols are different family.

- n-butyl alcohol and sec-butyl alcohol are position isomers of each other.

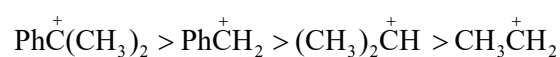
$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$: Butan-1-ol (n-butyl alcohol)

$\text{CH}_3\text{CH}_2\underset{\text{OH}}{\text{CH}}\text{CH}_3$: Butan-2-ol (sec-butyl alcohol)

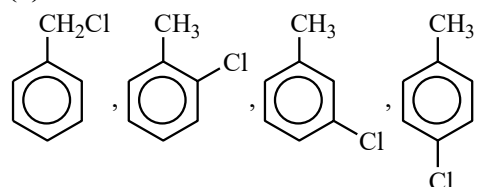
(NCERT Ed. 2025-26 11th Part-II Page No. 270)

51. (3)

Stability order:

(NCERT Ed. 2025-26 11th Part-II Page No. 275)

52. (2)

(NCERT Ed. 2025-26 11th Part-II Page No. 270)

53. (3)

- -I effect increases the acidic nature of compound.

- Acidic nature:

(NCERT Ed. 2025-26 11th Part-II Page No. 274)

54. (1)

Stability order:

Anti > Gauche > Partially eclipsed > Fully eclipsed

(NCERT Ed. 2025-26 11th Part-II Page No. 305)

55. (4)

Electromeric effect is a temporary effect

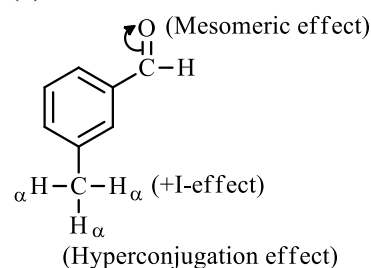
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56. (4)

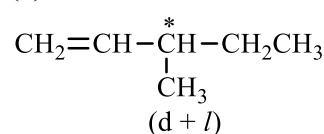
$\text{CH}_3\text{—}\overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}}\text{=CH}_2$ does not exhibit geometrical isomerism.

(NCERT Ed. 2025-26 11th Part-II Page No. 308)

57. (3)

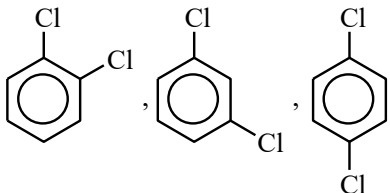
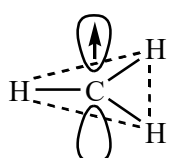
(NCERT Ed. 2025-26 11th Part-II Page No. 278)

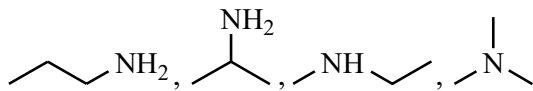
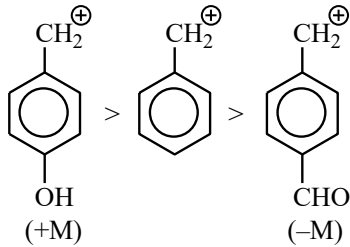
58. (1)

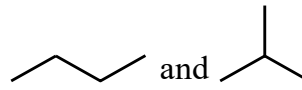
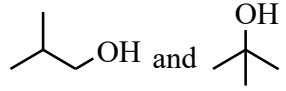
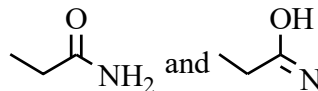
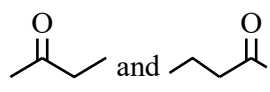
(NCERT Ed. 2025-26 12th Part-II Page No. 176)

59. (2)

-  : anti-aromatic
-  : aromatic
-  : aromatic
-  : anti-aromatic

- (NCERT Ed. 2025-26 11th Part-II Page No. 321)
60. (1)
Cis-trans isomers are not convertible to each other at room temperature.
(NCERT Ed. 2025-26 11th Part-II Page No. 308)
61. (3)
Stability order:
 $\text{HC}\equiv\text{C}^- > \text{CH}_2=\bar{\text{C}}\text{H} > \bar{\text{C}}\text{H}_3 > \text{CH}_3-\bar{\text{C}}\text{H}_2$
(NCERT Ed. 2025-26 11th Part-II Page No. 274)
62. (3)
Total number of conformers of ethane are infinite because there is almost free rotation about C–C bond in ethane.
(NCERT Ed. 2025-26 11th Part-II Page No. 305)
63. (1)
 $\text{CH}_2=\text{CH}-\dot{\text{C}}\text{H}_2 > \text{C}_6\text{H}_5-\dot{\text{C}}\text{H}_2 > \text{CH}_3-\dot{\text{C}}\text{H}_2 > \text{CH}_2=\dot{\text{C}}\text{H}$
(NCERT Ed. 2025-26 11th Part-II Page No. 278)
64. (2)

(NCERT Ed. 2025-26 11th Part-II Page No. 270)
65. (2)
 -  : sp^2 (trigonal planar)
 - $\text{CH}_3^\cdot$ is an electron deficient species but not behave as an electrophile
(NCERT Ed. 2025-26 11th Part-II Page No. 272)
66. (1)
Geometrical Isomers = $2^n = 2^2 = 4$
(NCERT Ed. 2025-26 11th Part-II Page No. 308)
67. (3)
Electromeric effect is observed in compounds containing π -bond.
(NCERT Ed. 2025-26 11th Part-II Page No. 277)
68. (3)
Acidic nature:
 $\text{H}_2\text{O} > \text{C}_2\text{H}_5\text{OH} > \text{CH}_3\text{COCH}_3 > \text{C}_6\text{H}_6$
(NCERT Ed. 2025-26 11th Part-II Page No. 274)

69. (1)
More the double bond character between C and Cl due to resonance, lesser will be C–Cl bond length.
(NCERT Ed. 2025-26 11th Part-II Page No. 276)
70. (3)

(NCERT Ed. 2025-26 11th Part-II Page No. 270)
71. (1)
 - Hyperconjugation involves overlapping of σ -p orbitals.
 - Hyperconjugation is also known as no bond resonance.
(NCERT Ed. 2025-26 11th Part-II Page No. 277)
72. (4)
 - $\text{CH}_3\text{CH}_2\text{CHO}$: Carbonyl compound
 - -OH: Cyclic alcohol
 - $\text{CH}_2=\text{CH}-\text{O}-\text{CH}_3$: Unsaturated ether
(NCERT Ed. 2025-26 11th Part-II Page No. 270)
73. (1)
Stability order:


(NCERT Ed. 2025-26 11th Part-II Page No. 275)
74. (2)
 -  : Chain isomers
 -  : Position isomers
 -  : Tautomers
 -  : Functional isomers
(NCERT Ed. 2025-26 11th Part-II Page No. 270)
75. (4)
Hydrocarbons having π -electrons behave as nucleophiles
(NCERT Ed. 2025-26 11th Part-II Page No. 273)

76. (1)
-
- (keto) (enol) (aromatic) (most stable)
- (NCERT Ed. 2025-26 11th Part-II Page No. 317)

77. (3)
-
- (2°) benzylic (more stable)
- (NCERT Ed. 2025-26 11th Part-II Page No. 272)

78. (1)
-
- cis trans
- (NCERT Ed. 2025-26 12th Part-II Page No. 176)

79. (3)
-
- is more stable than
- because octet of each carbon is complete in
- (NCERT Ed. 2025-26 11th Part-II Page No. 275)

80. (4)
-
- pentan-2-one pentan-2-one
- (NCERT Ed. 2025-26 11th Part-II Page No. 270)

81. (1)
- Non-polar structures are more stable than polar structures.
 - More the distance between opposite charges, lesser will be stability.
- (NCERT Ed. 2025-26 11th Part-II Page No. 275)

82. (3)
-
- (NCERT Ed. 2025-26 12th Part-II Page No. 176)

83. (3)
-
- Equivalent resonating structures contribute more towards stability.
- (NCERT Ed. 2025-26 11th Part-II Page No. 275)

84. (2)
- 2-Methyl pentane and 3-methyl pentane are chain isomers of each other.
- (NCERT Ed. 2025-26 11th Part-II Page No. 270)

85. (1)
- | | |
|--|-----------|
| | +R effect |
| $\text{CH}_3-\text{CH}=\text{CH}_2 \xrightarrow{\text{H}^+} \text{CH}_3-\text{CH}^+-\text{CH}_3$ | +E effect |
| | -E effect |
| | -R effect |
- (NCERT Ed. 2025-26 11th Part-II Page No. 276, 277)

86. (2)
- C_7H_{16} has nine chain isomers.
-
- (NCERT Ed. 2025-26 11th Part-II Page No. 270)

87. (3)
- In gaseous phase with increase in the number of alkyl groups the basic nature of aliphatic amines increases.
- (NCERT Ed. 2025-26 11th Part-II Page No. 274)

88. (4)
- $\text{C}_2\text{H}_6\text{O}$: $\text{CH}_3\text{CH}_2\text{OH}$ (ethanol), CH_3OCH_3 (Dimethyl ether)
- (NCERT Ed. 2025-26 11th Part-II Page No. 270)

89. (1)
- Negative charge is not delocalised so in $\text{CH}_3-\text{CH}_2^-$, carbon is sp^3 hybridised.
- (NCERT Ed. 2025-26 11th Part-II Page No. 272)

90. (4)
- Butyne (C_4H_6) and (C_4H_8) have different molecular formula.
- (NCERT Ed. 2025-26 11th Part-II Page No. 270)

91. (3)
The algae reproduce by vegetative, asexual and sexual methods.

(NCERT Ed. 2025-26, 11th, Page No. 24)

92. (2)
The earliest systems of classification used only gross superficial morphological characters such as habit, colour, number and shape of leaves, etc. They were based mainly on vegetative characters or on the androecium structure (system given by Linnaeus). Such systems were artificial.

(NCERT Ed. 2025-26, 11th, Page No. 23)

93. (1)
Certain marine brown and red algae produce large amounts of hydrocolloids (water holding substances), e.g., algin (brown algae) and carrageen (red algae) which are used commercially. Species of *Sphagnum* (a moss), a moss, provide peat. In angiosperms, the seeds are enclosed in fruits.

(NCERT Ed. 2025-26, 11th, Page No. 26)

94. (4)
Bryophytes are also called amphibians of the plant kingdom because these plants can live in soil but are dependent on water for sexual reproduction. They usually occur in damp, humid and shaded localities.

(NCERT Ed. 2025-26, 11th, Page No. 29)

95. (3)
The gametophytes in pteridophytes require cool, damp, shady places to grow. Because of this specific restricted requirement and the need for water for fertilisation, the spread of living pteridophytes is limited and restricted to narrow geographical regions.

(NCERT Ed. 2025-26, 11th, Page No. 32)

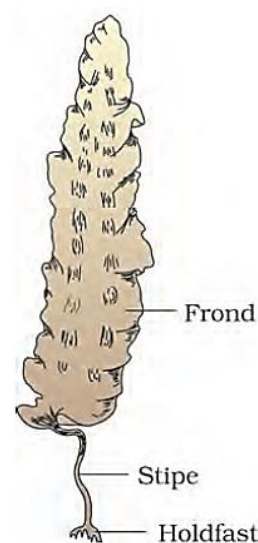
96. (4)
Unlike bryophytes and pteridophytes, in gymnosperms, the male and the female gametophytes do not have an independent free-living existence. They remain within the sporangia retained on the sporophytes. The gymnosperms are heterosporous; they produce haploid microspores and megaspores.

(NCERT Ed. 2025-26, 11th, Page No. 33)

97. (1)
The predominant stage of the life cycle of a moss is the gametophyte which consists of two stages. The first stage is the protonema stage, which develops directly from a spore. It is a creeping, green, branched and frequently filamentous stage. The second stage is the leafy stage, which develops from the secondary protonema as a lateral bud. This stage bears the sex organs. In sexual reproduction, the sex organs antheridia and archegonia are produced at the apex of the leafy shoots. After fertilisation, the zygote develops into a sporophyte, consisting of a foot, seta and capsule. The capsule contains spores. Spores are formed after meiosis.

(NCERT Ed. 2025-26, 11th, Page No. 30)

98. (2)
The plant body in brown algae is usually attached to the substratum by a holdfast, and has a stalk, the stipe and leaf like photosynthetic organ – the frond.



Brown alga: *Laminaria*

(NCERT Ed. 2025-26, 11th, Page No. 27)

99. (3)
Asexual reproduction in green algae is by flagellated zoospores produced in zoosporangia. Asexual reproduction in most brown algae is by biflagellate zoospores that are pear-shaped and have two unequal laterally attached flagella. The gametes in them are pyriform (pear-shaped) and bear two laterally attached flagella. The red algae reproduce asexually by non-motile spores and sexually by non-motile gametes.

(NCERT Ed. 2025-26, 11th, Page No. 27)

100. (2)

Floridean starch is the characteristic storage polysaccharide of red algae. Green algae store their food as starch. In brown algae, Food is stored as complex carbohydrates, which may be in the form of laminarin or mannitol.

(NCERT Ed. 2025-26, 11th, Page No. 27)

101. (1)

The leaves in pteridophyta are small (microphylls) as in *Selaginella* or large (macrophylls) as in ferns.

(NCERT Ed. 2025-26, 11th, Page No. 30)

102. (2)

Unlike the gymnosperms where the ovules are naked, in the angiosperms or flowering plants, the pollen grains and ovules are developed in specialised structures called flowers. The angiosperms are an exceptionally large group of plants occurring in wide range of habitats.

(NCERT Ed. 2025-26, 11th, Page No. 34)

103. (3)

Asexual reproduction in liverworts takes place by fragmentation of thalli, or by the formation of specialised structures called gemmae (sing. gemma). Gemmae are green, multicellular, asexual buds, which develop in small receptacles called gemma cups located on the thalli.

(NCERT Ed. 2025-26, 11th, Page No. 29)

104. (4)

The pteridophytes are further classified into four classes: Psilopsida (*Psilotum*); Lycopsida (*Selaginella*, *Lycopodium*), Sphenopsida (*Equisetum*) and Pteropsida (*Dryopteris*, *Pteris*, *Adiantum*).

(NCERT Ed. 2025-26, 11th, Page No. 32)

105. (2)

Fusion of two gametes dissimilar in size, as in species of *Eudorina* is termed as anisogamy. Fusion between one large, non-motile (static) female gamete and a smaller, motile male gamete is termed oogamy.

(NCERT Ed. 2025-26, 11th, Page No. 24)

106. (1)

At least a half of the total carbon dioxide fixation on earth is carried out by algae through photosynthesis.

(NCERT Ed. 2025-26, 11th, Page No. 26)

107. (1)

The main plant body of the bryophyte is haploid. It produces gametes, hence is called a gametophyte. During sexual reproduction in mosses, the sex organs antheridia and archegonia are produced at the apex of the leafy shoots. The mosses have an elaborate mechanism of spore dispersal. Zygotes of bryophytes do not undergo reduction division immediately. They produce a multicellular body called a sporophyte.

(NCERT Ed. 2025-26, 11th, Page No. 30)

108. (4)

The stems in gymnosperms may be unbranched (*Cycas*) or branched (*Pinus*, *Cedrus*). Angiosperms are called the flowering plants.

(NCERT Ed. 2025-26, 11th, Page No. 32)

109. (3)

Selaginella possesses vascular tissues.

(NCERT Ed. 2025-26, 11th, Page No. 30)

110. (2)

Called amphibians of plant kingdom - Bryophytes

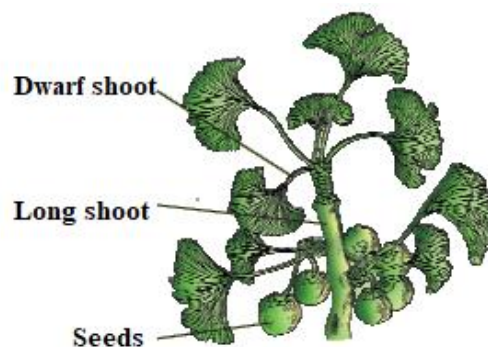
Include horsetails - Pteridophytes

Main plant body is thalloid - Algae

(NCERT Ed. 2025-26, 11th, Page No. 29)

111. (2)

Chara is a green alga. Green algae are characterized by having chlorophyll *a* and *b* as their main photosynthetic pigments. *Wolffia* is the smallest flowering plant. *Marchantia* is a genus of liverworts. The thallus is dorsiventral and closely appressed to the substratum like moist soil or rocks. *Ginkgo* is a unique gymnosperm tree. It is characterized by possessing two types of shoots: long, fast-growing shoots with spaced-out leaves, and short, stubby spur shoots where leaves are clustered.



Ginkgo

(NCERT Ed. 2025-26, 11th, Page No. 34)

112. (2)

The plant shown in the question is *Salvinia* (a petridophyte). Genera like *Selaginella* and *Salvinia* which produce two kinds of spores, macro (large) and micro (small) spores, are known as heterosporous. The megaspores and microspores germinate and give rise to female and male gametophytes, respectively. The female gametophytes in these plants are retained on the parent sporophytes for variable periods. The development of the zygotes into young embryos take place within the female gametophytes. This event is a precursor to the seed habit considered an important step in evolution.

(NCERT Ed. 2025-26, 11th, Page No. 32)

113. (4)

Fusion between one large, non-motile (static) female gamete and a smaller, motile male gamete is termed oogamous sexual reproduction, e.g., *Volvox*, *Fucus*.

(NCERT Ed. 2025-26, 11th, Page No. 24)

114. (1)

Natural classification systems were based on natural affinities among the organisms and consider, not only the external features, but also internal features. At present phylogenetic classification systems based on evolutionary relationships between the various organisms are acceptable.

(NCERT Ed. 2025-26, 11th, Page No. 23)

115. (3)

The microspores develop into a male gametophytic generation which is highly reduced and is confined to only a limited number of cells. This reduced gametophyte is called a pollen grain. The development of pollen grains take place within the microsporangia.

(NCERT Ed. 2025-26, 11th, Page No. 33)

116. (2)

The sporophyte in bryophytes is not free-living but attached to the photosynthetic gametophyte and derives nourishment from it. Zygotes do not undergo reduction division immediately. They produce a multicellular body called a sporophyte.

(NCERT Ed. 2025-26, 11th, Page No. 29)

117. (3)

In conifers, the needle-like leaves reduce the surface area. Their thick cuticle and sunken stomata also help to reduce water loss.

(NCERT Ed. 2025-26, 11th, Page No. 32)

118. (3)

Numerical Taxonomy which is now easily carried out using computers is based on all observable characteristics. Number and codes are assigned to all the characters and the data are then processed. In this way each character is given equal importance and at the same time hundreds of characters can be considered. Chemotaxonomy, that uses the chemical constituents of the plant to resolve confusions, is also used by taxonomists these days.

(NCERT Ed. 2025-26, 11th, Page No. 24)

119. (2)

Most of the members of Chlorophyceae have one or more storage bodies called pyrenoids located in the chloroplasts. Pyrenoids contain protein besides starch. Gemma cups are specialized cup-like structures that contain gemmae, which are small multicellular bodies used for asexual reproduction. These are a characteristic feature of certain liverworts. Prothallus is the small, free-living gametophyte stage in the life cycle of pteridophytes. Naked seeds are the defining characteristic of gymnosperms.

(NCERT Ed. 2025-26, 11th, Page No. 32)

120. (2)

In gymnosperms, the pollen tube carrying the male gametes grows towards archegonia in the ovules and discharge their contents near the mouth of the archegonia.

(NCERT Ed. 2025-26, 11th, Page No. 32)

121. (1)

Chlorella, a unicellular alga rich in proteins, is used as food supplement even by space travellers.

(NCERT Ed. 2025-26, 11th, Page No. 26)

122. (3)

At present phylogenetic classification systems based on evolutionary relationships between the various organisms are acceptable. This assumes that organisms belonging to the same taxa have a common ancestor. Linnaeus proposed an artificial system of classification. Bentham and Hooker proposed a natural system of classification.

(NCERT Ed. 2025-26, 11th, Page No. 24)

123. (1) After fertilisation in bryophytes, the zygote develops into a sporophyte, consisting of a foot, seta and capsule. Since mosses form dense mats on the soil, they reduce the impact of falling rain and prevent soil erosion. (NCERT Ed. 2025-26, 11th, Page No. 29) 124. (3) Kelps are marine algae. (NCERT Ed. 2025-26, 11th, Page No. 26) 125. (2) The brown algae possess possess chlorophyll <i>a</i>, <i>c</i>, carotenoids and xanthophylls. The vegetative cells have a cellulosic wall usually covered on the outside by a gelatinous coating of algin. (NCERT Ed. 2025-26, 11th, Page No. 26) 126. (3) In gymnosperms, one of the megaspores enclosed within the megasporangium develops into a multicellular female gametophyte that bears two or more archegonia or female sex organs. The multicellular female gametophyte is also retained within megasporangium. (NCERT Ed. 2025-26, 11th, Page No. 33) 127. (1) The angiosperms are divided into two classes : the dicotyledons and the monocotyledons. Depending on the type of pigment possessed and the type of stored food, algae are classified into three classes, namely Chlorophyceae, Phaeophyceae and Rhodophyceae. The pteridophytes are further classified into four classes: Psilopsida (<i>Psilotum</i>); Lycopsidea (<i>Selaginella</i>, <i>Lycopodium</i>), Sphenopsida (<i>Equisetum</i>) and Pteropsida (<i>Dryopteris</i>, <i>Pteris</i>, <i>Adiantum</i>). (NCERT Ed. 2025-26, 11th, Page No. 32) 128. (4) In bryophytes the dominant phase in the life cycle is the gametophytic plant body (haploid). However, in pteridophytes, the main plant body is a sporophyte (diploid). (NCERT Ed. 2025-26, 11th, Page No. 30)	129. (3) The reproduce asexually by non-motile spores and sexually by non-motile gametes. Sexual reproduction is oogamous and accompanied by complex post fertilisation developments. The common members are: <i>Polysiphonia</i>, <i>Porphyra</i>, <i>Gracilaria</i> and <i>Gelidium</i>. A defining characteristic of red algae is the complete absence of flagella throughout their entire life cycle. (NCERT Ed. 2025-26, 11th, Page No. 28) 130. (2) The rhizoids in mosses like <i>Funaria</i> are typically branched and multicellular, forming a network that helps anchor the plant and absorb water and minerals. (NCERT Ed. 2025-26, 11th, Page No. 30) 131. (3) A natural system of classification for flowering plants was given by George Bentham and Joseph Dalton Hooker. (NCERT Ed. 2025-26, 11th, Page No. 24) 132. (1) The function of air bladder present in the thallus of <i>Fucus</i> is to provide buoyancy. Holdfast helps in attachment to substratum. <i>Fucus</i> is a non-vascular alga that absorbs water and nutrients directly from the surrounding seawater across its entire surface. Photosynthetic pigments, such as chlorophyll and fucoxanthin, are located within the chloroplasts inside the cells of the thallus, not in the air bladders. (NCERT Ed. 2025-26, 11th, Page No. 25) 133. (1) Green algae usually have a rigid cell wall made of an inner layer of cellulose and an outer layer of pectose. (NCERT Ed. 2025-26, 11th, Page No. 26) 134. (2) The food in red algae is stored as floridean starch which is very similar to amylopectin and glycogen in structure. (NCERT Ed. 2025-26, 11th, Page No. 27) 135. (4) The female sex organ in bryophytes called archegonium is flask-shaped and produces a single egg. (NCERT Ed. 2025-26, 11th, Page No. 29)
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136. (2)

The drug is a competitive antagonist, meaning it blocks the normal action of GnRH at its receptors on the anterior pituitary. Since the primary function of GnRH is to stimulate the synthesis and release of LH and FSH, blocking this stimulation will lead to a rapid decline in their secretion.

(NCERT Ed. 2025-26, 11th, Page No. 240)

137. (2)

The hypothalamic-hypophyseal portal system is a network of capillaries that directly connects the hypothalamus to the anterior pituitary (adenohypophysis). Its specific function is to transport the releasing hormones (like GnRH, TRH) and inhibiting hormones (like somatostatin) from the hypothalamus to the anterior pituitary, allowing for precise and efficient regulation of its hormone secretion. It does not transport hormones to the posterior pituitary or from the pituitary to the rest of the body.

(NCERT Ed. 2025-26, 11th, Page No. 240)

138. (3)

TSH from the anterior pituitary stimulates the thyroid gland to produce and release thyroxine (T_4) and triiodothyronine (T_3), which regulate metabolism.

Prolactin promotes milk production in mammary glands.

Growth hormone mainly promotes tissue growth and metabolism; it does not directly stimulate thyroid hormone secretion.

ACTH stimulates the adrenal cortex to release glucocorticoids (e.g., cortisol).

(NCERT Ed. 2025-26, 11th, Page No. 241)

139. (1)

The nervous system uses neurons for fast, direct signaling.

The endocrine system uses hormones transported by blood, resulting in slower, widespread, and longer-lasting effects.

The "All-or-None" principle applies to nerve impulse generation. Hormonal responses are graded and depend on the concentration of the hormone and the number of available receptors.

The neural and endocrine systems are intricately linked and work together to regulate physiology (e.g., the hypothalamus controls the pituitary gland).

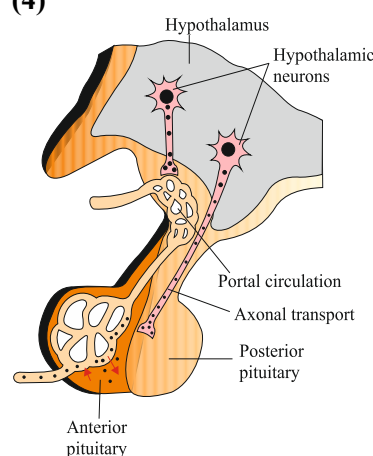
(NCERT Ed. 2025-26, 11th, Page No. 239)

140. (1)

List-I	List-II
Sella tursica	A bony depression in the sphenoid bone housing the pituitary
Pars distalis	Region responsible for synthesizing the majority of pituitary hormones, including tropic hormones
Pars intermedia	Almost merged with the pars distalis in humans, secretes MSH
Neurohypophysis	Releases hormones synthesized in hypothalamic neuron

(NCERT Ed. 2025-26, 11th, Page No. 241)

141. (4)

(NCERT Ed. 2025-26, 11th, Page No. 241)

142. (2)

The LH surge induces the rupture of the Graafian follicle, leading to ovulation. After ovulation, LH is also responsible for the formation and maintenance of the corpus luteum, which then secretes progesterone. However, the function described in reason (formation of corpus luteum) is a consequence of ovulation, not the reason why the LH surge causes ovulation itself.

(NCERT Ed. 2025-26, 11th, Page No. 241)

143. (1)

TSH and ACTH release is stimulated by TRH and CRH, respectively. Since these releasing hormones travel via the portal system, severing it would lead to a decrease in TSH and ACTH.

ADH is produced in the hypothalamus but travels to the posterior pituitary via axons, not the portal system. Since the axons are intact, its release mechanism would be largely unaffected.

Growth Hormone secretion is regulated by both a releasing hormone (GHRH) and an inhibiting hormone (Somatostatin), both of which use the portal system. Without either signal, its secretion would likely decrease and become non-responsive, not increase in an unregulated fashion.

(NCERT Ed. 2025-26, 11th, Page No. 240)

144. (3)

Hypothalamus synthesizes and releases releasing hormones (e.g., GnRH, TRH, CRH, GHRH) into the portal blood. These hormones regulate the secretion of hormones from the anterior pituitary (pars distalis).

Posterior pituitary (Neurohypophysis) only releases hormones (ADH and Oxytocin) that are synthesized in the hypothalamus. These hormones are transported via axons to the posterior pituitary and released into blood.

Anterior Pituitary synthesizes and releases hormones like GH, TSH, ACTH, LH, FSH, and prolactin. It is regulated by hypothalamic releasing hormones.

(NCERT Ed. 2025-26, 11th, Page No. 241)

145. (2)

Islets of Langerhans are the endocrine portion of the pancreas.

They secrete insulin, glucagon, somatostatin, and pancreatic polypeptide.

Though highly functional, they form only about 1–2% of total pancreatic tissue.

The rest (~98–99%) is exocrine, producing digestive enzymes.

(NCERT Ed. 2025-26, 11th, Page No. 245)

146. (4)

Glucagon, secreted by pancreatic α -cells, raises blood glucose mainly by acting on liver cells.

It stimulates:

Glycogenolysis → breakdown of glycogen to glucose.

Gluconeogenesis → synthesis of glucose from non-carbohydrate precursors (e.g., amino acids, lactate, glycerol).

(NCERT Ed. 2025-26, 11th, Page No. 245)

147. (3)

If the cause is a deficiency of ADH (central diabetes insipidus), administering synthetic ADH will provide the missing hormone. The kidneys are functional and will respond by increasing water reabsorption, leading to a significant decrease in urine output.

(NCERT Ed. 2025-26, 11th, Page No. 242)

148. (4)

Cortisol (a glucocorticoid) increases blood glucose by stimulating gluconeogenesis, reducing glucose uptake in peripheral tissues, and promoting protein catabolism for glucose synthesis.

Adrenaline (epinephrine) increases blood glucose via glycogenolysis in liver and skeletal muscle, and stimulates lipolysis for energy during stress.

(NCERT Ed. 2025-26, 11th, Page No. 245)

149. (3)

Secretin is secreted by S cells in the duodenum in response to acidic chyme (low pH). It stimulates the pancreatic duct cells to secrete a bicarbonate-rich juice to neutralize acid.

Cholecystokinin (CCK) is secreted by I cells in the duodenum and jejunum in response to fatty acids and partially digested proteins. It stimulates acinar cells of the pancreas to release enzyme-rich juice for digestion.

(NCERT Ed. 2025-26, 11th, Page No. 247)

150. (4)

Thymosins are hormones secreted by the thymus gland.

Thymosins do help immune function, promoting T-cell maturation and indirectly enhancing antibody production

(NCERT Ed. 2025-26, 11th, Page No. 243)

151. (4)

The thymus gland is located in the upper thoracic region, behind the sternum. It is large and most active during childhood but atrophies after puberty. Its main function is the maturation of T-lymphocytes for cell-mediated immunity.

Thymosins, peptide hormones secreted by the thymus, are crucial for this maturation and immune competence.

(NCERT Ed. 2025-26, 11th, Page No. 243)

152. (3)

Growth factors (e.g., epidermal growth factor, platelet-derived growth factor) are generally paracrine (acting on nearby cells) or autocrine (acting on the same cell) signaling molecules. This local action is important for processes like tissue repair, regeneration, and cell proliferation.

Classical endocrine hormones (e.g., thyroxine) are secreted into the bloodstream and act on distant target organs.

(NCERT Ed. 2025-26, 11th, Page No. 247)

153. (2)

List-I	List-II
Testosterone	Stimulates muscular growth
Estrogen	Development of female secondary sex characters
Progesterone	Supports pregnancy
LH (in males)	Stimulates Leydig cells

(NCERT Ed. 2025-26, 11th, Page No. 246)

154. (3)

Testis functions both as endocrine gland (testosterone secretion), primary sex organ (produces sperm).

Testosterone induces deepening of voice (low pitch is normal; testosterone makes it deeper).

The pituitary gland is attached to the hypothalamus via the infundibular stalk.

Androgens regulate growth of pubic, facial, and axillary hair during puberty.

(NCERT Ed. 2025-26, 11th, Page No. 246)

155. (2)

T₃ and T₄ maintain the basal metabolic rate and also have secondary effects on water and electrolyte balance through their influence on metabolism and renal function.

While iodine is essential for T₃ and T₄ synthesis, TCT synthesis does not require iodine because it is a protein hormone unrelated to iodination.

(NCERT Ed. 2025-26, 11th, Page No. 243)

156. (3)

In untreated diabetes mellitus, lack of effective insulin action prevents glucose uptake by cells.

The body switches to fatty acid oxidation as the primary energy source.

Excess acetyl-CoA from β -oxidation is converted to form ketone bodies (acetoacetic acid, β -hydroxybutyric acid, acetone).

These acids accumulate in the blood (ketosis), lowering pH (ketoacidosis), and can appear in urine (ketonuria).

(NCERT Ed. 2025-26, 11th, Page No. 246)

157. (1)

The parathyroid glands secrete PTH, the primary hormone for raising blood calcium. Its removal will cause a swift and severe drop in blood Ca²⁺ levels.

Calcium is essential for stabilizing nerve and muscle membranes. Severe hypocalcemia increases neuromuscular excitability, leading to spontaneous muscle contractions, spasms, and convulsions, a condition known as hypocalcemic tetany. This is a life-threatening emergency.

While TCT will be absent, TCT's role in humans is minor compared to PTH. Its absence will not cause hypercalcemia, especially when PTH, the main hypercalcemic hormone, is also gone. The effect of PTH loss will overwhelmingly dominate.

Removal of thyroid hormones will cause the metabolic rate to decrease, not increase.

(NCERT Ed. 2025-26, 11th, Page No. 243)

158. (3)

The adrenal medulla secretes catecholamines, mainly epinephrine (adrenaline) and norepinephrine (noradrenaline), under sympathetic nervous system control.

A tumor of the adrenal medulla (e.g., pheochromocytoma) causes excess catecholamine secretion, leading to symptoms such as paroxysmal hypertension, palpitations, excessive sweating, headaches, and anxiety.

(NCERT Ed. 2025-26, 11th, Page No. 244)

159. (2)

Gastric Inhibitory Peptide (GIP) is also known as Glucose-dependent Insulinotropic Peptide, it is a hormone secreted by K-cells in the duodenum and jejunum.

It has two main physiological roles:

- Inhibits gastric functions i.e. decreases gastric motility and acid secretion.
- Stimulates insulin release i.e. acts as an incretin hormone, enhancing insulin secretion from pancreatic β -cells in response to oral glucose.

(NCERT Ed. 2025-26, 11th, Page No. 247)

160. (1)

The thymus gland degenerates with age, leading to a reduced output of naive T-cells. This decline in T-cell function (immunosenescence) is a primary reason for the weakened immune system in the elderly, explaining both susceptibility to infections and poor vaccine efficacy.

Cortisol, the long-term stress hormone, raises blood sugar (via gluconeogenesis) and can cause hypertension. Critically, it also suppresses the immune system. Therefore, chronic stress leading to high cortisol levels can explain both the metabolic symptoms and can contribute to the weakened immune state.

Aldosterone is secreted from the adrenal cortex (not medulla) and it raises (not lowers) blood pressure.

Catecholamines are for short-term stress and do not have a primary role in destroying lymphocytes to cause long-term immunosuppression in the way cortisol does.

(NCERT Ed. 2025-26, 11th, Page No. 243)

161. (4)

List-I	List-II
Zona glomerulosa	Mineralocorticoids (e.g., Aldosterone)
Zona fasciculata	Glucocorticoids (e.g., Cortisol)
Zona reticularis	Gonadocorticoids (e.g., Androgens)

(NCERT Ed. 2025-26, 11th, Page No. 245)

162. (4)

The RAAS is activated by low blood pressure and acts to raise it (via vasoconstriction from angiotensin II and sodium/water retention from aldosterone).

ANF is activated by high blood pressure and acts to lower it.

One of the key mechanisms by which ANF lowers blood pressure is by inhibiting the RAAS. It suppresses the release of renin from the kidneys, which in turn reduces the production of angiotensin II and aldosterone. This prevents the blood pressure-raising effects of the RAAS, complementing ANF's direct vasodilatory and natriuretic actions.

(NCERT Ed. 2025-26, 11th, Page No. 247)

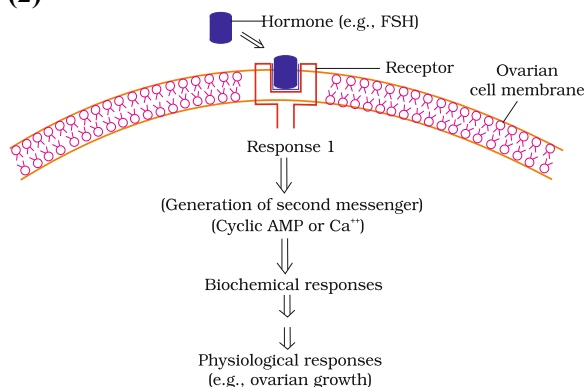
163. (2)

The hypothalamus is the basal (ventral) part of the diencephalon of the forebrain.

The hypothalamus plays a key role in controlling/integrating autonomic nervous system activity and endocrine functions (e.g., via the pituitary).

(NCERT Ed. 2025-26, 11th, Page No. 240)

164. (2)



(NCERT Ed. 2025-26, 11th, Page No. 248)

165. (1)

Steroid hormones (e.g., cortisol, aldosterone) are lipid-soluble, diffuse across the plasma membrane, and bind to intracellular receptors (cytoplasmic or nuclear). The hormone–receptor complex then modulates gene transcription.

Thymosin is a peptide hormone secreted by the thymus, crucial for the maturation and differentiation of T-lymphocytes, which are central to cell-mediated immunity.

(NCERT Ed. 2025-26, 11th, Page No. 248)

166. (1)

The secretion of melatonin is stimulated by darkness and inhibited by light. Therefore, its levels are low during the day and high during the night. This nightly peak in melatonin is what helps induce and maintain sleep.

(NCERT Ed. 2025-26, 11th, Page No. 242)

167. (3)

The nervous system is fast and targeted, but its "wiring" does not connect to every single cell. However, metabolic and other cellular functions in all tissues need constant, widespread regulation. This creates a physiological need for a system that can communicate with all cells, even those not directly touched by a neuron.

Endocrine system uses chemical messengers (hormones) that travel via the circulatory system to reach virtually all cells in the body, providing slow, long-lasting, and widespread coordination.

(NCERT Ed. 2025-26, 11th, Page No. 239)

168. (2)

High altitude means lower partial pressure of oxygen, leading to systemic hypoxia. The body's primary long-term adaptation to this is to increase its ability to carry oxygen. The kidney detects this chronic hypoxia and responds by upregulating the production and secretion of erythropoietin (EPO). EPO stimulates the bone marrow to produce more red blood cells (RBCs). An increased number of RBCs (a higher hematocrit) means the blood can carry more oxygen per unit volume, which compensates for the lower oxygen availability in the air. This is the principle behind altitude training for endurance athletes.

(NCERT Ed. 2025-26, 11th, Page No. 247)

169. (3)

- GIP and Secretin: Both inhibit HCl secretion. Their effects are synergistic, not antagonistic.
- CCK and GIP: Both inhibit HCl secretion. Their effects are synergistic, not antagonistic.
- Gastrin and Somatostatin: Gastrin stimulates HCl secretion, while Somatostatin inhibits HCl secretion. These two have direct, opposing effects and form a crucial local feedback loop to control stomach acid. This is a classic antagonistic pair.
- Secretin and CCK: Both inhibit HCl secretion. Their effects are synergistic, not antagonistic.

(NCERT Ed. 2025-26, 11th, Page No. 247)

170. (3)

Glucagon is a peptide hormone. Being water-soluble, it cannot cross the cell membrane and must bind to specific membrane-bound receptors on the surface of liver cells.

Cortisol is a steroid hormone. Being lipid-soluble, it can cross the cell membrane and bind to intracellular receptors within the cytoplasm or nucleus.

(NCERT Ed. 2025-26, 11th, Page No. 248)

171. (3)

Adrenal cortex secretes cortisol, aldosterone, and androgens.

Cortisol and especially aldosterone are vital for life: Aldosterone maintains electrolyte balance and blood pressure.

Hence, removal of adrenal cortex causes death unless treated with hormone replacement therapy.

Endocrine glands are called ductless gland. They secrete their products (hormones) directly into the bloodstream or the surrounding interstitial fluid, from where they are transported to target cells. They lack the ducts that exocrine glands (like salivary or sweat glands) use to carry their secretions to a specific surface.

(NCERT Ed. 2025-26, 11th, Page No. 245)

172. (2)

List-I	List-II
Peptide hormone	Insulin
Steroid hormone	Estradiol
Iodothyronine	Thyroxine
Amino-acid derivative	Adrenaline

(NCERT Ed. 2025-26, 11th, Page No. 248)

173. (2)

Classical hormones are secreted by well-defined endocrine glands (e.g., pituitary, thyroid).

Modern endocrinology includes hormones from non-classical sources (e.g., ANF from the heart), so not only ductless glands.

Paracrine and autocrine messengers from non-endocrine tissues (e.g., growth factors, nitric oxide) are now classified under the broader hormone concept. While glucose levels are a powerful signal (e.g., they signal the pancreas to release insulin), glucose itself is not a hormone. Hormones are molecules produced by cells specifically for the purpose of signaling. Glucose is primarily a nutrient and energy source. Therefore, it is not included under the definition of a hormone.

(NCERT Ed. 2025-26, 11th, Page No. 239)

174. (2)

Hormones synthesized and secreted by pars distalis are: TSH, LH, FSH, ACTH, GH, and Prolactin → 6 total

(NCERT Ed. 2025-26, 11th, Page No. 241)

175. (1)

List-I	List-II
Graves' disease	Enlargement of the thyroid gland, protrusion of the eyeballs, increased basal metabolic rate, and weight loss
Cretinism	Mental retardation, low intelligence quotient, abnormal skin, deaf-mutism
Hyperthyroidism	Hypersecretion of thyroid hormones, leading to increased BMR
Hypothyroidism	A condition characterized by decreased secretion of thyroid hormones in an adult

(NCERT Ed. 2025-26, 11th, Page No. 242)

176. (3)

Thyroid gland consists of spherical follicles and the intervening stromal tissue.

The follicular cells synthesise the iodothyronine hormones (T₄ and T₃). The hormone thyrocalcitonin (TCT or calcitonin) is secreted by the parafollicular cells (C-cells) located in the stromal tissue between the follicles, not by the follicular cells themselves.

(NCERT Ed. 2025-26, 11th, Page No. 242)

177. (3)

Insulin's function is to lower blood glucose. An overdose will cause an excessive drop in blood glucose levels, a condition known as hypoglycemia. The brain is critically dependent on a constant supply of glucose for its function. Severe or profound hypoglycemia can impair brain function, leading to confusion, seizures, loss of consciousness, and a state known as a hypoglycemic (or diabetic) coma, which can be fatal if not treated immediately with glucose.

(NCERT Ed. 2025-26, 11th, Page No. 246)

178. (1)

In the developing foetus, thyroid hormones (primarily from the mother during early gestation) are essential for normal brain development, especially the maturation and differentiation of neurons and proper myelination.

Maternal hypothyroidism or congenital hypothyroidism can lead to cretinism, characterized by severe and irreversible mental retardation if untreated within the first weeks of life.

(NCERT Ed. 2025-26, 11th, Page No. 242)

179. (2)

Although the pituitary gland regulates many endocrine organs, it is itself regulated by the hypothalamus, which secretes releasing and inhibiting hormones (like TRH, GnRH, CRH, GHRH, and somatostatin). This shows that the ultimate control of the endocrine system lies in the hypothalamic–pituitary axis.

(NCERT Ed. 2025-26, 11th, Page No. 240)

**180. (2)**

List-I	List-II
Gigantism	Hypersecretion of GH in children
Acromegaly	Hypersecretion of GH in adults
Diabetes Insipidus	Deficiency of ADH
Addison's Disease	Hyposecretion of adrenal cortex hormones

(NCERT Ed. 2025-26, 11th, Page No. 241)



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